
Trigonometric Identities and Equations — Illegal Cancellation and Lost Solutions

Cancelling is the fastest way to lose a solution you were never told about. Three rules cover almost every case.

- **Never divide both sides by an expression containing the variable.** Move everything to one side and *factor* instead: the zero-product property then keeps every branch.
- **Only factors of a product cancel.** A term of a sum may never be cancelled against a denominator.
- **Squaring both sides can create solutions that are not there.** Every candidate must be checked in the *original* equation.
- In each problem: name the error in words first, then show corrected work. Unless told otherwise, solve on $0^\circ \leq \theta < 360^\circ$.

1. Find and fix. Solve $2 \sin \theta \cos \theta = \sin \theta$ on $0^\circ \leq \theta < 360^\circ$.

Rafi divides both sides by $\sin \theta$, gets $\cos \theta = \frac{1}{2}$, and reports $\theta = 60^\circ, 300^\circ$.

- (a) Name what Rafi's division assumed without saying so.
(b) Solve the equation correctly and list every solution.

Answer: _____

2. Find and fix. Simplify $\frac{\cos^2 x - 1}{\cos x - 1}$.

Bea cancels the $\cos^2 x$ against the $\cos x$ and the 1 against the 1, and writes $\cos x$.

(a) Simplify the expression correctly by factoring the numerator.

(b) Show Bea's form is wrong by evaluating both at a value where $\cos x = 0.5$, and give the correct value there.

Answer: _____

3. Find and fix. Solve $2 \cos^2 \theta = \cos \theta$ on $0^\circ \leq \theta < 360^\circ$.

Nils divides both sides by $\cos \theta$ and reports $\theta = 60^\circ, 300^\circ$.

(a) Write the equation in factored form with 0 on one side.

(b) List every solution, and say which ones Nils's division threw away.

Answer: _____

4. Find and fix. A student simplifies $\frac{\sin^2 x + \cos x}{\cos x}$ to $\sin^2 x + 1$.

- (a) Explain why the $\cos x$ in the numerator cannot be cancelled against the denominator.
- (b) Test both forms at a value where $\sin^2 x = 0.75$ and $\cos x = 0.5$. Give the correct value there.

Answer: _____

5. Find and fix. Solve $4 \sin^2 \theta = 1$ on $0^\circ \leq \theta < 360^\circ$.

Tam takes the square root of both sides, writes $\sin \theta = \frac{1}{2}$, and reports $\theta = 30^\circ, 150^\circ$.

- (a) Say what taking a square root leaves out.
- (b) List every solution on the interval.

Answer: _____

6. Find and fix. Solve $\sin \theta - \cos \theta = 1$ on $0^\circ \leq \theta < 360^\circ$.

Ola squares both sides, simplifies to $\sin \theta \cos \theta = 0$, and reports all four of $\theta = 0^\circ, 90^\circ, 180^\circ, 270^\circ$.

(a) Ola's algebra is correct but her answer is not. Say what squaring did to the problem.

(b) Test each candidate in the original equation and list the solutions that actually work.

Answer: _____

7. Find and fix. Solve $\cos 2\theta = \cos \theta$ on $0^\circ \leq \theta < 360^\circ$.

Yuki "cancels the cosine from both sides" to get $2\theta = \theta$, and reports $\theta = 0^\circ$ only.

(a) Explain why equal cosine values do not force equal angles.

(b) Use a double-angle identity to write the equation with 0 on one side, factor it, and list every solution.

Answer: _____

8. Explain.

- (a) Explain why dividing both sides of an equation by a trigonometric expression can lose solutions, and state what the division silently assumes.
- (b) Explain why a term of a sum may never be cancelled against a denominator, even though a factor of a product may. Give a numerical example that makes the difference visible.
- (c) Explain why squaring both sides forces you to check every candidate, and contrast that with what dividing does. One operation adds solutions and the other removes them — say which is which and why.